NoVa data center asset **Scope 1 and 2 emission** calculator

Target user:
Asset-level data center owners and operators

Use case:
Measure **Scopes 1 and 2** for operating, near-stabilized/stabilized data center assets (for better operational decarbonization planning)

- The primary use case applies to **core, core+, and value-add data-center investments (which means more for funds/REITs/AMs, rather than developers)**, where emissions reductions are driven by operational efficiency, power procurement, and equipment optimization during the investment hold period.

Reminder on what are scope 1 and 2:

Scope 1:

- Direct GHG emissions from sources that are owned or controlled by the data center asset

Scope 2:

- GHG emissions indirectly resulting from the generation of electricity, heat or steam purchased by the data center asset

Hypothesis – not really, just for reference and sanity check:

For a typical **grid-connected** data center:

Emission Source	**Relative Share**
Scope 2 – IT Electricity	~60–75%
Scope 2 – Cooling & Overhead	~20–30%
Scope 1 – Diesel Testing	~1–5% (normal ops)
Scope 1 – Gas Combustion	<5% (if applicable)

In most stabilized U.S. assets: **Scope 2 = 90%+ of operational carbon footprint**

Flow Summary

Carbon Baseline

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Normalized Carbon Intensity

Carbon intensity normalization:

- kg CO_2 / MWh
- kg CO_2 / MW IT load
- kg CO_2 / revenue proxy
- per Rack
- per PUE-adjusted output

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Emission Driver Decomposition

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Marginal Abatement Insights

Marginal abatement insight:

For each lever, show:

- Δ CO_2 per year
- Rough CapEx band (low / medium / high)
- Operational complexity (low / medium / high)

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Rough CapEx + Operational Complexity Overlay (low / medium / high)

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Action Prioritization

[Calculator Inputs] Data Center Asset Scope 1 and 2 emissions breakdown

Scope 1: direct emissions on-site

- Calculator SCOPE 1 emission INPUTS:
  - Annual diesel fuel consumed (gallons)
  - Annual natural gas consumed (therms or MMBtu)
  - Total refrigerant charge (kg) per equipment type
  - Annual leakage rate (%)
  - Refrigerant type (to determine GWP)
- FORMULAE
  - Overarching formula : Total Scope 1 (tCO_2e) = Diesel Emissions + Natural Gas Emissions + Refrigerant Emissions
  - DIESEL : Diesel Emissions (kg CO_2) = Diesel Consumed (gallons) × 10.21 kg CO_2/gallon
  - ON-SITE GAS: Natural Gas Emissions (kg CO_2) = Natural Gas (therms) × 5.3 kg CO_2/therm
  - REFRIGERANT leakage: Refrigerant Emissions (kg CO_2e) = Σ [Refrigerant Charge (kg) × Leak Rate (%) × GWP]
    - GWP below
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Refrigerant	Chemical Name	GWP (100-yr)
R-410A	HFC blend	2,088
R-134a	HFC-134a	1,430
R-407C	HFC blend	1,774
R-32	HFC-32	675
R-22 (legacy)	HCFC-22	1,810
R-744	CO_2	1
R-717	Ammonia	0

    - https://www.ipcc.ch/report/ar5/wg1/

Scope 2: purchased electricity (2 main uses: IT loads and cooling systems)

- Calculator Scope 2 emission inputs:
  - Total IT load (MW)
  - PUE
  - Grid emission factor (269 kg CO_2 per MWh - for post code 20147; source: https://www.epa.gov/egrid/power-profiler#/SRVC)
  - % of renewable energy powering the data center
- Formulae:

- Absolute Scope 2 emission = IT MW * 8760 * PUE * grid emission factor * (1 - % of renewable)
- Scope 2 carbon Intensity per IT MWh = PUE * grid emission factor * (1 - % of renewable)

* assume IT load full utilization rate
* assume steady full-year operation
* assume no battery storage used onsite for peak shaving or carbon shifting

* assume renewables have zero Scope 2 emissions

Energy Source	**Operational Emissions (Scope 2 EF)**
Solar	~0 kg CO_2/MWh
Wind	~0 kg CO_2/MWh
Nuclear	~0 kg CO_2/MWh
Hydropower	~0 kg CO_2/MWh
Geothermal	~0 kg CO_2/MWh
Renewable PPA / REC-backed	~0 kg CO_2/MWh

Decompose carbon intensity:

$$\text{Carbon per IT MWh} = \underbrace{PUE}_{Operational} \times \underbrace{GridEF}_{Location} \times \underbrace{(1 - Renewable\%)}_{Procurement}$$

So your tool can calculate:

- % impact from PUE
- % impact from grid carbon
- % reduction potential from renewables

1️⃣ **If Carbon Intensity is High**

Ask:

Is PUE high?

Define benchmarks:

- Best-in-class PUE: 1.2
- Market average: 1.4–1.6
- Legacy: >1.6

If:

PUE > 1.5
→ Flag as operational inefficiency driver

If YES (i.e. PUE is high) → Operational inefficiency issue
Recommendation:

- Upgrade cooling architecture (**CapEx band: Medium; Operational complexity: Medium**)
- Improve airflow containment (**CapEx band: Low; Operational complexity: Low → Medium**)
- Retrofit to liquid cooling for high-density AI racks (**CapEx band: High; Operational complexity: High**)
- Replace legacy chillers (**CapEx band: Medium → High; Operational complexity: High**)
- Optimize load balancing (**CapEx band: Low; Operational complexity: Low**)

If user improves PUE from current to benchmark:

$$\Delta Scope2 = IT \times 8760 \times (PUE_{old} - PUE_{new}) \times GridEF$$

That gives:

- Absolute tCO_2 reduction
- % reduction

Your consulting output becomes:

"Improving PUE from 1.6 to 1.3 would reduce emissions by 18%."

Marginal Carbon Sensitivity of PUE reduction

Because emissions scale linearly:

$$\frac{d(Scope2)}{dPUE} = IT \times 8760 \times GridEF$$

That tells investor:

"How much does each 0.1 PUE reduction save?"

Example output:

"Each 0.1 reduction in PUE reduces emissions by 3,500 tCO_2 annually."

Translation:

Focus on facility efficiency capex.

Is Grid EF high (carbon-heavy region)? – include NoVa specific insights

If:

NoVa GridEF > national median
→ Flag as grid-driven exposure

If YES (i.e. GridEF is high) → Location / procurement issue
Recommendation:

- Procure renewable PPAs (**CapEx Band: Low → Medium**; **Operational Complexity: Medium**)
- Enter green tariff programs (**CapEx Band: Low; Operational Complexity: Low → Medium**)
- Acquire RECs (short-term solution; accounting only) (**CapEx Band: Very Low**; **Operational Complexity: Low**)
- Consider new campus development in lower-carbon region (**CapEx Band: High; Operational Complexity: High**)
- Build onsite solar (**CapEx Band: Medium; Operational Complexity: Medium**)
- Build or Co-Locate With Nuclear / Dedicated Generation (**CapEx Band: Very High**; **Operational Complexity: Very High**)

Since we are assuming zero Scope 2 emissions for renewables:

> If Renewable% = 20%, then 80% remains grid-exposed.
>
> So you can show:
>
> "Increasing renewable coverage to 100% would reduce market-based Scope 2 by 80%."

Translation:

Power sourcing strategy matters more than engineering.

2️⃣ If PUE is Already Efficient (<1.3)

Then:

Upgrading IT structure (servers) may not reduce Scope 2 unless:

- It reduces IT load for same compute output
- Or improves workload efficiency per MW

Important nuance:

If you upgrade to more powerful GPUs:

- IT MW may increase
- Scope 2 increases
- But compute output per MW may increase even more

So raw emissions might go up while **carbon per compute falls.**

- Same Compute, Lower IT MW → Scope 2 decreases (**CapEx Band: Medium; Operational Complexity: Medium**)
- More Powerful GPUs, Higher IT MW → Scope 2 increases (**CapEx Band: High; Operational Complexity: High**)
- Same IT MW, More Compute Output from better chip architecture → Scope 2 unchanged but carbon per compute decreases (**CapEx Band: Low → Medium; Operational Complexity: Low → Medium**)

That's why benchmarking normalized intensity matters.

At the end of your calculator, show:

A) Absolute Emissions

(tCO_2/year)

B) Carbon per MW IT

(tCO_2 per MW IT per year)

C) Carbon per MWh IT

(PUE × Grid EF)

Make a quadrant chart to classify assets based on structural and operational features:

Quadrant 4: High PUE, Low Electricity EF

→ Efficiency issue.

Quadrant 2: Low PUE, High Electricity EF

→ Location/available energy source issue.

Quadrant 3: Both High

→ Energy efficiency and energy source dual risk.

Quadrant 1: Both Low

→ Efficient asset.

PUE

Formula

$$\text{PUE} = \frac{\text{Total Facility Energy}}{\text{IT Equipment Energy}}$$

Where:

- **Total Facility Energy** = everything (servers + cooling + lighting + UPS losses + fans + etc.)
- **IT Equipment Energy** = just servers, storage, networking

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What the number means

- **PUE = 2.0**
  For every 1 kW used by servers, another 1 kW is used for overhead (cooling, power losses).
  → 50% efficiency to IT load.
- **PUE = 1.5**
  1 kW IT → 0.5 kW overhead.
  → 67% efficiency to IT load.
- **PUE = 1.2** (hyperscaler-level)
  Very efficient facility.
- **PUE = 1.0**
  Theoretical perfect efficiency (impossible in practice).

Why this calculator can be useful to investors?

Comparability across assets (this is where it gets real)

Once standardized, your calculator allows:

- Asset A vs Asset B
- Same data center before vs after an intervention
- Portfolio-level rollups

This is *extremely* useful for:

- Private equity portfolios
- Infrastructure funds
- Operators managing 10–100+ facilities

Regulatory + reporting readiness

Scope 1 & 2 are:

- Mandatory under most disclosure regimes
- The least controversial, most auditable scopes

Your tool helps with:

- ESG reporting
- LP transparency
- Internal governance
- Avoiding “we’ll fix it later” chaos

Slide Deck Planning

Give an example calculation in the slide deck

At the end of the calculator, give a scale and relevant recommendation for each grade of the emission per power unit (maybe)

- Heatbar showing which area does the emission come from